

Frontend technical stack

Native SwiftUI client on `origin/development` at commit `88f147e`. The UI commits on this branch are by HONG Shangjing (`23259124@life.hkbu.edu.hk`). This report covers screen structure, interaction logic, and the platform split. It describes the code on that commit; later plans are not treated as shipped behavior.

1. Stack

There is no web client. The Mac app and the iOS app each host the same `WorkspaceView`. Platform differences are `#if os` gates around files, local compute, the 3D preview, and PDF sharing.

Layer	Choice	Role in the interface
Language and UI	Swift 6, SwiftUI	Every screen. Deployment floor is macOS 14 and iOS 17.
State	Observation	WorkspaceStore owns navigation and the open project. ProjectSession owns selection, the dirty flag, and validation. Both are <code>@MainActor</code> and <code>@Observable</code> .
2D plan	SwiftUI Canvas	Top-down room. Tap selects. Drag moves. Hit testing and drawing live in <code>TopDownRoomView</code> .
3D preview	RealityKit RealityView	Schematic room on macOS 15 and later. macOS 14 and iOS show a written fallback.
Design system	SimuDesignSystem	Accent tint, rounded headings, page intros, a small floor-plan mark, and a shared empty state. No third-party UI kit.
Files	fileImporter, NSSavePanel, ShareLink	Open a <code>.simunow</code> package, save it, and share a PDF.
Copy	Simplified Chinese literals	Labels, empty states, and errors are hardcoded. There is no string catalog.

Table 1. Runtime stack behind the screens. Source: SimuKit on `origin/development`.

`SimuKit` is one Swift package. The app targets depend only on `SimuWorkspace`. That module depends on `SimuCore`, `SimuSimulation`, `SimuVisualization`, `SimuReporting`, and `SimuDesignSystem`. Views do not calculate loads. `SimuVisualization` draws geometry and does not invent a temperature or airflow field. `SimuReporting` writes PDF bytes; the report page decides when export is allowed.

`RoomStyle` sets the palette: ink `#173C49`, air `#146A73`, mist `#EAF3F4`, slate `#596E76`, caution `#B66F16`. Light mode tints controls with air; dark mode uses a lighter teal (rgb 0.33, 0.72, 0.75). Headings use a rounded system face. Status pairs an SF Symbol with words, so color is not the only signal. VoiceOver labels exist on sidebar rows, numeric fields (including the unit), the wizard progress bar, and the 3D scene. The floor-plan mark is labeled as a diagram, not a result.

`SimulationViewoport` is still an empty-state placeholder and is not mounted by the workspace. The live pictures of the room are `TopDownRoomView` and `RoomPreview3D`.

2. Information architecture

`WorkspaceView` has two modes. With no open project it shows `HomeView`. With a project it shows a `NavigationSplitView`: sidebar, center, and inspector. Closing the project clears the session and resets `RunStore`.

Home

The home list is the only screen before a project exists. Each action opens a different path into the same store.

Action	Result	Interaction rule
Create from my room	Five-step wizard sheet	Next stays disabled until the name, the three dimensions, and every opening are valid. Finish builds one rectangular room and a baseline scenario with empty HVAC.
Office template	Opens immediately	Four seats in a 5×4 m room. Hong Kong October climate is already filled. The far wall is north until the user changes it.
Classroom template	Opens immediately	Twelve seats in an 8×6 m room. Same climate preset and the same north default.
Open a saved room	Folder importer	Security-scoped access, then ProjectPackageCodec. An unsupported version tells the user to import a v1 draft or update the app.
Import old draft	JSON importer	ProjectMigrator. Migration notes appear in an alert. HVAC and ventilation still need filling.
Recent projects	Up to ten paths	Stored in UserDefaults. A sandbox failure asks the user to pick the file again through the open panel.

Table 2. Home-screen entry paths.

Wizard

Step	Question shown to the user	Gate
1 / 5	What kind of room is this?	Non-empty name. Space type: office, classroom, home, or public.
2 / 5	How large is the room?	Width, depth, and height in metres, all positive. North angle is optional.
3 / 5	Where are the doors and windows?	Optional. Offset, size, and wall must keep the opening on its surface.
4 / 5	Which walls face outdoors?	Unchecked faces are adiabatic. U-value and SHGC sit in a professional disclosure.
5 / 5	Confirm the room	Climate, date, and time zone use the HKU October preset. HVAC and ventilation stay unknown on purpose.

Table 3. Room-creation wizard. Progress is announced to VoiceOver as step N of 5.

Wall names in the wizard and the sidebar are screen-relative: left wall, right wall, near wall, far wall, floor, ceiling. The floor-plan mark says these names are not compass directions. A disclosure explains that the north angle is measured clockwise from the far wall (+Y) to true north.

Project shell

The sidebar is 180 – 220 points wide. The inspector is 260 – 380 points. The center title follows the selected destination.

Column	Contents	What a click does
Sidebar	Four destinations, then the room tree: walls, openings, furniture, the air conditioner, seats, people, equipment heat, the remote setpoint, weather, and tariffs.	A destination button switches the center. An entity row sets session.selection. Openings are named by wall. A setpoint is named by its device.
Center	布置房间 (place the room), 方案对比 (compare), 用电估算 (estimate), 导出报告 (export).	The toolbar saves (a star appears while dirty), opens the validation sheet, and closes the project. A dirty close asks to save, discard, or cancel.
Inspector	A grouped form dispatched by EntitySelection.	An empty selection explains how to pick an object. A validation jump sets focusedSetting and scrolls that anchor to the center.

Table 4. Three-column project shell.

The first save on the Mac uses `NSSavePanel` and writes a `.simunow` package. On iOS the same save writes into the app Documents directory, with no panel.

3. Room editing

The center of 布置房间 has a segmented control: 俯视编辑 (plan) and 3D 预览 (preview). The plan is the editing surface on every platform. The 3D view is a geometry preview that can also move objects. A caption under the control states the current gesture: on the plan, tap to edit and drag to move; in 3D, drag an object along the floor and drag empty space to orbit.

Plan view

TopDownRoomView scales one rectangular room into the canvas with a 60-point margin. Screen Y is domain depth, flipped so the far wall is at the top. If the room is not a known rectangle with known dimensions, the canvas is replaced by an unavailable state.

- Furniture is a gray box. A seat is a green 12-point dot with its name. An air conditioner is a blue rounded rectangle.
- Each port draws a 14-point arrow. Supply is orange. Return is purple.
- The selected object gets an accent outline.
- A legend under the canvas repeats those three colors in words.

A tap hit-tests seats, devices, and furniture centers inside 24 points and keeps the nearest. A drag of at least 4 points selects the object under the pointer and moves it. Movement snaps to 0.05 m and is clamped inside the room. Seat sample points and device ports move with their parent. Furniture keeps its size and moves its origin. The plan writes into the project on every drag update through `session.mutate`, which marks the file dirty and revalidates.

Three buttons sit under the plan, wrapping onto a second line when the column is narrow. **Add seat** places a seat at the room center, adds a person, and adds sample heights at 0.1, 0.6, and 1.1 m. **Add furniture** places a $1.2 \times 0.6 \times 0.75$ m desk. **Add air conditioner** places one split unit near the far wall, with supply, return, and a setpoint. Those added numbers are stored as explicit assumptions, including a note that they are placeholders. Supply temperature stays unknown until the user fills it. A caption button says whether the project can be estimated and opens the issue list.

3D preview

RoomPreview3D uses RealityKit on macOS 15 and later. macOS 14 shows a note that the plan can still complete the model. iOS and iPadOS show a note that the plan is the layout tool on those devices. The scene is geometry only. The accessibility label says it contains no airflow or temperature field.

Meshes are schematic: floor, translucent walls, desk, chair, person, monitor, wall-mounted unit, door, and window. People and monitors are display stand-ins taken from occupant and equipment positions. The mesh size is not a measured body or device. Supply and return arrows are parented to the unit so they move with it.

- Dragging an object slides it on a horizontal plane at that object's current height. A desk stays on the floor. A wall unit keeps its mounting height.
- Dragging empty space orbits the camera. Elevation is limited to about $0.03 - 1.45$ radians.
- A pinch scales distance between $0.35 \times$ and $3 \times$ the suggested camera distance ($2.2 \times$ the longest room side, plus 0.5 m).
- Reset restores azimuth $-\pi/5$, elevation $\pi/6$, and distance factor 1.
- The scene rebuilds only when the layout fingerprint changes and no drag is active, so orbiting does not reconstruct the room.
- The domain point is written once, when the gesture ends, and only if it changed.

Shared move rules

Rule	Behavior
Snap	0.05 m on X, Y, and Z before clamping.
Clamp	The object and its attached points stay inside the rectangular room. If the span is larger than the room, the anchor is pinned to the lower bound.
Height	A plan drag keeps Z. A 3D drag zeros the vertical delta, so height does not change.
Attachments	A seat moves its sample points. A device moves its ports. Furniture keeps its size. Equipment is a point.
Commit	The plan writes during the drag. The 3D view writes once, on gesture end.
Coordinates	Domain is metres, right-handed, Z-up. Display is Apple Y-up: x stays, domain z becomes y, and domain y becomes -z. The conversion is <code>CoordinateTransform</code> .

Table 5. Rules shared by the plan and the 3D preview, implemented in `RoomPreviewDrag` and `RoomMotion`.

Inspector

With nothing selected, the form says to click an air conditioner, a seat, or a piece of furniture, or to use the sidebar for walls, weather, and price. A selection that no longer exists is cleared. Changing the selection clears the focused validation anchor.

`ParameterRow` is the only control for a physical quantity. A known value shows the number and a fixed unit, plus a disclosure for its source: measured, manufacturer, preset, scan, user, or assumed. Measured, manufacturer, and preset sources require a reference. An assumption requires a note. Choosing “leave this blank” clears the number and requires a reason. Filling a blank value needs a parseable finite number and an explicit Confirm. The control does not insert a silent default. The known-value binding keeps the previous source and uncertainty when the number changes.

Supply-air temperature and the remote setpoint are different forms. The device form says the supply temperature is the air leaving the unit. The control form says it is the number on the remote. Cooling capacity and electrical power are separate fields, with a note that they come from the nameplate. Professional details — body-heat split, discharge direction as a unit vector, weather-file hash — stay inside disclosures. A validation jump expands the north-angle or weather section when the anchor path contains those fields, waits 200 ms, and scrolls the anchor to the center.

The issue sheet has two status rows: whether the project is internally valid, and whether estimation can start. Each issue shows a plain-language field name, a short action, and a button that jumps to the setting when the path can be mapped. The raw code and path sit in a disclosure.

4. Estimate, compare, and export

These three pages read stored runs. They do not recompute the physics. A result is eligible for comparison and for the report only when the run completed, quality passed, and the input hash still matches the scenario.

Estimate

If the Python worker is not configured, the Mac page says this machine cannot estimate yet and hides the setup command inside a disclosure. The room can still be edited and saved. On iOS the same page says estimation runs on a Mac; the phone can edit the room and read results that already exist. `WorkspaceStore.defaultRunClient` never starts a local process on iOS.

When a worker is available, 估算当前方案 stays disabled until input preparation passes. A second button opens the issue list, and each issue can send the user back to that field, switching the center to the room editor and selecting the entity. The page states that the estimate is a whole-room average and does not provide a temperature or an airflow at each seat.

- Status is an icon plus a word: running, completed, failed, cancelled, or interrupted.

- Every row carries a “whole-room average” badge.
- Quality reads 检查通过, 检查未通过, or 尚未检查.
- Freshness reads 最新, or 待重算 when the scenario hash has changed since the run.
- A running row shows a spinner and a Cancel button.
- The disclosure holds the metric table, the assumptions used, errors, the run id, the event log, and the run directory.
- A missing metric renders as 暂无法估算 and discloses a reason. It is not shown as zero.

Compare

The page intro asks which setting fits, and tells the user to duplicate a scenario, change the temperature, and estimate each copy. A caption states that the comparison covers electricity and cooling capacity, and that seat comfort is not judged.

- Each scenario row can be made current, or duplicated as a candidate with the same entity ids and a new scenario id.
- “Try 1 ° C higher and lower” adds setpoint copies inside the allowed range and does not change the scenario the user is on. The button is disabled when the current setpoint has no value.
- Three cards cover how to choose on electricity, whether capacity is enough, and seat comfort. Each card can expand its evidence and its limits.
- The comfort card always says seats are not evaluated.
- Only scenarios that share weather, occupancy, and schedule are ranked together. Others are named in an exclusion line.
- Ineligible runs are listed with the reason they were left out.
- A detailed metric grid, including an eight-character run id, sits behind a disclosure.
- The cost section lists currency, tariffs, and quotes. It says a missing price is not turned into a cost, and that the figure is for the selected day.

Export

The report page stays an empty state until **ReportBuilder** can assemble content from at least one eligible run. The empty state points the user back to 用电估算. The preview lists the included scenarios and, inside a disclosure, a short run id, input-hash prefix, and quality. It states that the document is one day’s average estimate and cannot be used to judge comfort at each seat. Export on the Mac uses a save panel named {project}-报告.pdf. On iOS the file is written to the temporary directory and a ShareLink appears after it succeeds.

5. Platform split

Action	macOS	iOS and iPadOS
Plan, wizard, inspector	Shared SwiftUI views.	The same views.
3D preview	RealityKit on macOS 15+. A written note on macOS 14.	A written note. The plan remains the layout tool.
Local L0 estimate	Python worker when configured. Otherwise a setup disclosure.	Blocked in the UI. Existing results can still be read.
First save	NSSavePanel, .simunow package.	App Documents directory, no panel.
PDF	Save panel.	Generate a temporary file, then ShareLink.
Window	Default 1280 × 800, minimum 900 × 600.	System window. The split view collapses with the size class.

Table 6. Interactions that differ by platform. The shell is one `NavigationSplitView`; there is no separate iPhone navigation design.

6. Limits of this interface

The following are properties of the screens on this commit, not missing documentation.

- One rectangular room. An unknown room shape is kept as data and is not drawn or dragged.
- The 3D preview and the plan show geometry. They do not show a CFD field, a comfort map, or an animation of cooldown.
- Results shown in the UI are L0 whole-room averages. The pages say so on the run row, the comparison intro, and the report preview.
- The comfort recommendation card states that seat comfort cannot be evaluated.
- A single simulated day is not presented as an annual saving. Cost is omitted when no price is stored.
- Copy is Simplified Chinese only.
- iPhone uses the system collapsed split view. There is no dedicated compact navigation.
- `SimulationViewport` remains an unmounted empty state reserved for a later results renderer.

Evidence: `Apps/SimuNowMac`, `Apps/SimuNowiOS`, and `Packages/SimuKit` sources `SimuWorkspace`, `SimuVisualization`, `SimuDesignSystem`, and `SimuReporting` on `origin/development` at 88f147e. Reviewed 3 October 2026.